

Project Proposal
in
Cloud Computing in Web Development 1

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Chapter I: Introduction

Project Overview

The Smart Gym Finance Manager is a cloud-based web application designed to help gym owners efficiently manage their financial operations, membership records, and business performance through intelligent analytics. Traditional gym management often relies on spreadsheets or manual bookkeeping, making it difficult to monitor income, expenses, membership renewals, and financial growth.

The proposed system centralizes all financial transactions into one platform while integrating Artificial Intelligence (AI) to generate revenue forecasts, predict membership cancellations, and detect unusual expenses. These intelligent features enable gym owners to make informed business decisions and improve operational efficiency. By providing automated financial reports, real-time dashboards, and predictive analytics, the system minimizes human error, reduces administrative workload, and increases profitability.

Project Scope

The Smart Gym Finance Manager focuses on financial and membership management for small to medium-sized fitness centers.

The system includes:

- Membership payment management
- Member renewal monitoring
- Revenue tracking
- Expense management
- Profit and Loss reporting
- Financial dashboard
- AI revenue forecasting
- Membership churn prediction
- Expense leak detection
- PDF and Excel report generation
- Administrator login and account management

The project excludes:

- Workout scheduling
- Personal trainer management
- Nutrition tracking
- Attendance through biometrics
- Online coaching
- Mobile application development

The system will be deployed as a cloud-based web application accessible through modern web browsers.

Key Features

1.1. Membership Payment Tracking

Records member payments including membership plans, payment status, due dates, and payment history.

1.2. Monthly Revenue Dashboard

Displays graphical summaries of monthly earnings, membership growth, expenses, and net income.

1.3. Expense Tracking

Allows administrators to record operational expenses including:

- Equipment purchases
- Utility bills
- Employee salaries
- Rent
- Maintenance costs

1.4. Profit and Loss Reports

Automatically calculates monthly and yearly profits and losses based on recorded income and expenditures.

1.5. Membership Renewal Monitoring

Tracks membership expiration dates and sends reminders for upcoming renewals.

1.6. Export Reports

Generate financial reports in:

- PDF
- Microsoft Excel

1.7. AI Revenue Forecast

Uses historical financial data, membership trends, seasonal changes, and renewal rates to predict future monthly income.

Example:

Current Members: 180

Expected Renewals: 160

Predicted New Members: 20

Forecasted Revenue:
₱145,000

1.8. Membership Churn Prediction

Uses attendance history and membership expiration dates to identify members likely to discontinue their subscriptions.

Example:

Member hasn't visited for 20 days.

Membership expires in 7 days.

Risk Level:
High Risk

1.9. Expense Leak Detector

Analyzes monthly operating expenses to identify unusual increases.

Example:

Average Utility Bill:
₱8,000

Current Utility Bill:
₱15,000

System Alert:

"Utility expenses increased by **87.5%** compared to the monthly average."

Target Client

The proposed system is intended for:

- Small fitness gyms
- Medium-sized fitness centers
- Independent gym owners
- Fitness club administrators

- Accounting personnel

These organizations require an affordable and intelligent solution to simplify financial management while improving profitability and operational efficiency.

Problem Statement

Many gym businesses continue to rely on manual bookkeeping, spreadsheets, or disconnected software for managing finances and memberships. These traditional methods often result in inaccurate financial records, delayed reporting, missed membership renewals, and poor decision-making due to the lack of real-time financial insights.

Additionally, gym owners have limited capability to anticipate future revenue, identify members who are likely to cancel their subscriptions, or detect abnormal increases in operating expenses. As a result, businesses may experience financial losses, reduced customer retention, and inefficient allocation of resources.

The Smart Gym Finance Manager addresses these challenges by integrating AI-driven financial analytics, predictive forecasting, and automated reporting into a centralized cloud-based platform that supports smarter financial planning and business management.

Sustainable Development Goal (SDG) Connection

This project supports the following United Nations Sustainable Development Goals:

SDG 8 – Decent Work and Economic Growth

The system promotes sustainable business growth by improving financial management, increasing operational efficiency, and supporting long-term profitability for fitness businesses.

SDG 9 – Industry, Innovation and Infrastructure

The integration of Artificial Intelligence and cloud computing encourages innovation in business operations while modernizing financial management practices.

SDG 12 – Responsible Consumption and Production

Expense monitoring and leak detection encourage responsible financial planning and efficient resource utilization.

Project Objectives

General Objective

To develop a cloud-based Smart Gym Finance Manager that improves financial management and business decision-making through AI-powered analytics and automated reporting.

Specific Objectives

The project aims to:

1. Develop a secure membership payment tracking system.
2. Automate monthly revenue and expense recording.
3. Generate Profit and Loss reports in real time.
4. Monitor membership renewal schedules.
5. Generate downloadable PDF and Excel reports.
6. Predict monthly revenue using AI forecasting techniques.
7. Identify members likely to cancel their memberships.
8. Detect abnormal expense increases through intelligent expense analysis.
9. Provide an interactive dashboard for monitoring financial performance.
10. Improve operational efficiency by reducing manual financial processes.

Chapter II: Review of Related Literature

his chapter presents the related literature and studies that support the development of the **Smart Gym Finance Manager**. The reviewed studies focus on cloud computing, financial management information systems, artificial intelligence, business intelligence, gym management systems, and predictive analytics. These studies provide valuable information about how technology improves financial operations, enhances decision-making, and automates business processes. The literature is divided into two sections: local studies conducted in the Philippines and international studies from other countries.

Local Studies

1. Bulaclac, J. R., Pastorfide, J. M. V., Herrera, N. G., & Tambio, C. N. C. (2025). Revolutionizing Gym Management: A QR-Enabled Cashless System for Smarter Payments and Tracking.

Bulaclac et al. (2025) developed a QR-enabled cashless payment and product monitoring system specifically designed for fitness centers. The researchers found that replacing manual payment methods with QR-based transactions significantly improved transaction speed, reduced administrative workload, and increased the accuracy of financial and inventory records. The study also reported a high usability score, indicating that both gym staff and members found the system easy to use.

2. Villanueva, M. C., Sevilla, S. A. S., & Verecio, R. L. (2025). Development of Olympic Fitness Gym Management System with Integrated QR Code Technology.

Villanueva et al. (2025) developed a gym management system that integrates QR code technology for attendance monitoring, membership management, and payment processing. The researchers found that automating gym operations reduced paperwork, improved transaction efficiency, and increased customer satisfaction through faster services. The study supports the proposed Smart Gym Finance Manager because it demonstrates that digital technologies improve gym administration. Unlike the referenced study, the proposed system also focuses on financial analytics by providing AI-powered forecasting, expense monitoring, and financial reporting to assist gym owners in making better business decisions.

3. Huang, S. (2025). Optimization of Intelligent Financial Management System Based on Blockchain and Internet of Things.

Huang (2025) proposed an intelligent financial management system that integrates blockchain technology and the Internet of Things to improve financial accuracy, efficiency, and security. The study concluded that intelligent financial systems simplify financial processes, reduce operational costs, and provide more accurate financial records for organizations. Although the study focuses on educational institutions, its findings are applicable to the proposed Smart Gym Finance Manager. Both systems utilize intelligent technologies to automate financial processes and improve decision-making through accurate financial information and modern digital tools.

4. Bulaclac, J. R., et al. (2025). QR-Based Payment Systems for Fitness Industry Digital Transformation.

The researchers emphasized that digital payment systems are becoming essential in modern fitness centers because they improve operational efficiency and customer convenience. The study found that automated payment systems reduce human error, enhance data security, and provide real-time monitoring of financial transactions. This study supports the membership payment tracking feature of the Smart Gym Finance Manager. Similar to the proposed system, it demonstrates that digital financial solutions improve business operations while reducing manual workloads for administrators.

5. Bulaclac, J. R., et al. (2025). Technology Acceptance of QR-Based Gym Management Systems.

The researchers evaluated the usability and acceptance of QR-based gym management systems using the Technology Acceptance Model (TAM) and the System Usability Scale (SUS). The study revealed that users perceived the system as useful, easy to learn, and reliable,

resulting in high acceptance among both gym staff and customers. The findings support the proposed Smart Gym Finance Manager because the success of any financial management system depends on user acceptance and ease of use. The proposed system adopts the same principle by providing a simple interface, automated financial processes, and intelligent analytics that are easy for gym administrators to understand and use.

6. Batuto, E. J. J. (2025). Artificial Intelligence-Enhanced Financial Management Information System for Church-Based Institutions: A Case Study of the Diocese of San Carlos.

Batuto (2025) developed an AI-enhanced Financial Management Information System to improve financial transparency and automate accounting processes within church institutions. The study integrated machine learning for anomaly detection and revenue forecasting, resulting in improved reporting accuracy and faster transaction processing. Users also rated the system highly for usability and reliability. This study is closely related to the proposed Smart Gym Finance Manager because both systems use Artificial Intelligence to automate financial management. The AI Revenue Forecast and Expense Leak Detector of the proposed system are based on similar concepts of predictive analytics and anomaly detection to help administrators make informed financial decisions.

7. Esteban, A. P., Piad, K., Tano, I., Espino, J., Lagman, A. C., Victoriano, J., & Mababa, J. (2024). Implementation of Digital Governance in the Philippine SUCs: Basis for an Enterprise-Level Information System Model.

Esteban et al. (2024) studied the implementation of enterprise-level information systems in Philippine State Universities and Colleges. The researchers found that centralized digital information systems improved organizational efficiency, data accessibility, and decision-making while reducing paperwork and redundant processes. Although the study focused on educational institutions, its findings are relevant to the Smart Gym Finance Manager because both systems rely on centralized cloud-based databases. The proposed project similarly aims to improve financial monitoring and administrative efficiency through an integrated web application.

8. Santiago, R. L., Villarica, M., & Bernardino, M. (2025). Budget and Financial Management Information System for Public Elementary Schools: Analytics and Predictive Insights for MOOE Allocation Using Linear Regression.

Santiago et al. (2025) developed a Budget and Financial Management Information System that integrates Artificial Intelligence and predictive analytics to automate budgeting and financial reporting for public schools. The study demonstrated that AI-based forecasting improved budget allocation and increased the accuracy of financial reports while reducing manual computation. The proposed Smart Gym Finance Manager shares the same objective of

improving financial management through predictive analytics. Both systems use intelligent technologies to analyze historical financial data and assist administrators in making better financial decisions.

9. Rodriguez, J. M. (2024). The AI, Blockchain, Cloud and Data (ABCD) Technology Integration in the Philippines: A Literature Review.

Rodriguez (2024) reviewed the adoption of Artificial Intelligence, Blockchain, Cloud Computing, and Data Analytics across different industries in the Philippines. The study concluded that these technologies significantly improve operational efficiency, security, and decision-making while highlighting cloud computing as an effective platform for managing business information systems. The findings support the proposed Smart Gym Finance Manager because the project is built using cloud computing and integrates AI-powered analytics. The study reinforces the importance of combining intelligent technologies to improve financial monitoring and business performance.

10. Bisnar, R., Atencio, M., Deriquito, E. J., Bueno, P., & Roldan, R. G. (2025). Financial Management System with Budgeting, Financial Forecasting and Planning with AI for Bestlink College of the Philippines.

Bisnar et al. (2025) developed an AI-powered Financial Management System capable of budgeting, forecasting, and financial planning. The researchers found that integrating AI into financial management improved reporting accuracy, minimized processing delays, and supported data-driven decision-making for administrators. This study strongly supports the Smart Gym Finance Manager because both systems integrate Artificial Intelligence into financial operations. Similar to the proposed project, the study demonstrates how AI forecasting and automated financial reporting can improve efficiency, planning, and overall financial management.

International Studies

1. Huang, S. (2025). Optimization of Intelligent Financial Management System Based on Blockchain and Internet of Things.

Huang (2025) developed an intelligent financial management system that integrates blockchain technology, the Internet of Things (IoT), and Artificial Intelligence to improve financial transparency and operational efficiency. The study found that intelligent financial systems significantly reduced manual work, increased the accuracy of financial records, and improved decision-making through automated financial analysis. This study is relevant to the proposed Smart Gym Finance Manager because both systems utilize intelligent technologies to automate financial processes. While Huang's study focuses on institutional financial management, the

proposed project applies similar AI-driven concepts to help gym owners forecast revenue, monitor expenses, and improve financial planning.

2. Goyal, N., & Kumar, P. (2024). Artificial Intelligence in Financial Management: Opportunities and Challenges.

Goyal and Kumar (2024) examined how Artificial Intelligence transforms financial management across different industries. The researchers found that AI technologies improve financial forecasting, detect unusual transactions, automate reporting, and assist managers in making strategic decisions. The study also emphasized that AI reduces human errors while increasing productivity. The findings strongly support the Smart Gym Finance Manager because the proposed system incorporates AI Revenue Forecasting and Expense Leak Detection. Similar to the study, the project aims to automate financial analysis and provide business owners with accurate financial insights.

3. Wirtz, B. W., Weyerer, J. C., & Geyer, C. (2022). Artificial Intelligence and the Public Sector: Applications and Challenges.

Wirtz, Weyerer, and Geyer (2022) explored various applications of Artificial Intelligence in organizational management. Their research found that AI improves data analysis, automates repetitive administrative tasks, and enables organizations to make faster and more accurate decisions based on large datasets. The study also highlighted the growing importance of predictive analytics in business management. Although the study focused on public administration, its findings are applicable to the Smart Gym Finance Manager. The proposed system also uses predictive analytics to estimate future revenue and identify members who are likely to discontinue their memberships.

4. IBM Corporation (2023). Artificial Intelligence for Financial Forecasting.

IBM discussed how Artificial Intelligence helps organizations analyze historical financial records to predict future revenue and improve budgeting strategies. The report explained that predictive analytics allows businesses to prepare for future financial conditions and minimize unexpected financial risks through data-driven planning. This supports the proposed Smart Gym Finance Manager because one of its primary features is AI Revenue Forecasting. Like IBM's approach, the proposed system analyzes historical membership payments, renewal rates, and seasonal trends to estimate future income and assist gym owners in making informed financial decisions.

5. Microsoft Corporation (2023). Power BI Financial Analytics and Business Intelligence.

Microsoft introduced Power BI as a business intelligence platform capable of transforming raw financial data into interactive dashboards and visual reports. The study explained that organizations using business intelligence dashboards can monitor revenues, expenses, and financial performance in real time, allowing faster and more informed decision-making. This study directly supports the Monthly Revenue Dashboard feature of the proposed Smart Gym Finance Manager. Similar to Power BI, the proposed system presents financial information through graphs, charts, and summarized reports, enabling gym administrators to easily evaluate business performance and identify financial trends.

6. Google Cloud. (2024). AI and Machine Learning Solutions for Business Intelligence.

Google Cloud (2024) discussed how Artificial Intelligence and machine learning help businesses analyze large amounts of financial data to identify trends, forecast future performance, and improve decision-making. The report explained that AI-powered business intelligence enables organizations to make proactive decisions instead of relying solely on historical reports. Businesses that adopted AI analytics experienced improved efficiency, better resource allocation, and more accurate financial planning. This study is closely related to the proposed Smart Gym Finance Manager because the system also uses AI to analyze financial records and predict future revenue. Similar to Google's approach, the proposed project transforms business data into meaningful insights that help gym owners improve financial performance and operational planning.

7. Microsoft Azure. (2024). Machine Learning for Predictive Analytics in Business.

Microsoft Azure (2024) explained how machine learning models are used to predict customer behavior, analyze business performance, and automate financial decision-making. The report highlighted that predictive analytics allows organizations to identify customer retention risks and forecast future business trends using historical data. This study supports the Membership Churn Prediction feature of the proposed Smart Gym Finance Manager. By analyzing attendance records and membership expiration dates, the proposed system predicts which members are likely to discontinue their subscriptions, allowing administrators to implement retention strategies before cancellations occur.

8. Amazon Web Services (AWS). (2023). Cloud Computing for Modern Business Applications.

Amazon Web Services (2023) discussed how cloud computing enables businesses to securely store, manage, and access information through internet-based platforms. The report emphasized that cloud-based systems reduce infrastructure costs, improve scalability, and provide secure data storage with automatic backup and recovery features. The findings support the proposed Smart Gym Finance Manager because the project is designed as a cloud-based web application. Similar to AWS cloud solutions, the proposed system enables gym owners to

securely access financial records, membership information, and reports from any location with an internet connection.

9. Oracle Corporation. (2024). Artificial Intelligence in Enterprise Financial Management.

Oracle (2024) explored the integration of Artificial Intelligence into enterprise financial management systems. The report concluded that AI significantly improves budgeting, expense monitoring, financial forecasting, and operational efficiency by automating repetitive financial tasks and identifying unusual financial patterns before they become serious problems. This study directly supports the Expense Leak Detector and AI Revenue Forecast modules of the Smart Gym Finance Manager. Both systems rely on intelligent data analysis to detect financial anomalies and provide recommendations that help administrators make informed financial decisions.

10. Deloitte. (2023). Finance 2025: Digital Transformation and Intelligent Financial Management.

Deloitte (2023) examined how digital transformation is changing financial management across different industries. The report found that organizations adopting cloud computing, Artificial Intelligence, and business analytics experienced faster financial reporting, improved decision-making, and better overall business performance. It also emphasized that intelligent financial systems are becoming essential tools for sustainable business growth. This study strongly supports the overall concept of the Smart Gym Finance Manager. The proposed project combines cloud computing, AI forecasting, financial reporting, and business analytics into a single web application that enables gym owners to efficiently manage finances while making strategic business decisions.

Corroborative Synthesis

The reviewed local and international studies consistently demonstrate that modern technologies such as cloud computing, Artificial Intelligence, machine learning, and business intelligence have significantly improved financial management across various industries. The local studies emphasized the growing adoption of intelligent financial systems in the Philippines, particularly in improving transaction processing, financial reporting, budgeting, and organizational efficiency. Likewise, the international studies highlighted how AI-powered analytics, cloud-based platforms, and predictive technologies enable organizations to automate financial operations, forecast future performance, monitor expenses, and make data-driven decisions.

A common theme among the reviewed literature is the importance of automation in reducing manual workloads and improving operational accuracy. Several studies showed that digital

financial systems reduce human errors, simplify reporting procedures, and provide administrators with real-time access to financial information. Another recurring finding is that Artificial Intelligence enhances decision-making through predictive analytics, allowing organizations to forecast revenue, identify financial risks, and improve customer retention. These findings strongly support the proposed Smart Gym Finance Manager, which integrates membership payment tracking, expense monitoring, profit and loss reporting, AI revenue forecasting, membership churn prediction, and expense leak detection into a cloud-based platform.

Although many existing studies focus on financial management systems, only a few specifically address the financial operations of fitness centers. Most available systems concentrate on membership management or digital payment processing without incorporating intelligent financial analytics. This research gap highlights the need for the proposed Smart Gym Finance Manager. By combining cloud computing with Artificial Intelligence, the proposed system provides a comprehensive solution that not only manages financial records but also assists gym owners in making strategic business decisions. The reviewed literature therefore establishes the relevance, feasibility, and potential contribution of the proposed project in improving financial management practices within the fitness industry.

Chapter III: Prototype

Low-Fidelity Prototype and Development Plan

The Smart Gym Finance Manager is composed of eight (8) integrated modules that work together to provide an efficient, intelligent, and cloud-based financial management platform for fitness centers. The system is designed to help gym owners and administrators monitor membership payments, manage operational expenses, generate financial reports, and utilize Artificial Intelligence for business forecasting and decision-making. Since the system is cloud-based, users can securely access their financial records and reports anytime and anywhere through an internet connection.

System Modules

Module 1: User Authentication and Account Management Module

Purpose:

This module manages administrator login, authentication, authorization, and profile management to ensure that only authorized users can access the system.

Key Features:

Administrator Login

Secure Logout

Password Encryption

Password Recovery

Profile Management

Role-Based Access Control

Inputs:

Username

Password

User Information

Outputs:

Authenticated User Session

User Profile

Access Permissions

Interaction with Other Modules:

This module serves as the gateway to all other modules. Users must successfully log in before accessing the dashboard, financial reports, membership records, and AI-powered analytics.

Module 2: Dashboard Module

Purpose:

Provides administrators with an overview of the gym's financial performance and membership statistics through interactive charts and summary cards.

Key Features:

Monthly Revenue Overview

Total Expenses

Net Profit

Active Members

Expiring Memberships

Revenue and Expense Graphs

AI Notifications

Inputs:

Membership Records

Expense Records

Financial Data

Outputs:

Dashboard Charts

Financial Summaries

Business Statistics

Interaction with Other Modules:

The dashboard gathers information from the Membership Management, Expense Management, Financial Reports, and AI modules to display real-time business performance.

Module 3: Membership Management Module

Purpose:

Allows administrators to manage member information, monitor memberships, and record payment transactions.

Key Features:

Add Member

Update Member Information

Membership Plans

Payment Recording

Membership Renewal Monitoring

Membership Expiration Tracking

Inputs:

Member Information

Membership Type

Payment Details

Outputs:

Updated Membership Records

Payment History

Membership Status

Interaction with Other Modules:

Membership information is used by the Dashboard, Financial Reports, AI Revenue Forecast, and Membership Churn Prediction modules.

Module 4: Expense Management Module

Purpose:

Records and manages all operational expenses of the fitness center.

Key Features:

Record Expenses

Edit Expenses

Delete Expenses

Expense Categorization

Monthly Expense Summary

Inputs:

Expense Name

Expense Category

Amount

Date

Outputs:

Updated Expense Records

Expense Summary Reports

Interaction with Other Modules:

Expense data is shared with the Dashboard, Financial Reports, and Expense Leak Detector module.

Module 5: Financial Reports Module

Purpose:

Generates automated financial reports to help administrators monitor the financial performance of the gym.

Key Features:

Revenue Reports

Expense Reports

Profit and Loss Statement

Monthly Reports

Annual Reports

Export to PDF

Export to Excel

Inputs:

Revenue Data

Expense Data

Membership Payments

Outputs:

Financial Reports

Downloadable PDF Files

Excel Reports

Interaction with Other Modules:

Collects financial information from the Membership Management and Expense Management modules and provides summarized reports through the Dashboard.

Module 6: AI Revenue Forecast Module

Purpose

Predicts the gym's expected monthly revenue using Artificial Intelligence based on historical financial and membership data.

Key Features:

Revenue Prediction

Membership Trend Analysis

Seasonal Trend Analysis

Financial Forecasting

Inputs:

Previous Revenue Records

Active Memberships

Membership Renewals

New Registrations

Outputs:

Predicted Monthly Revenue

Revenue Forecast Report

Interaction with Other Modules:

Receives financial information from the Membership Management and Financial Reports modules and displays forecast results on the Dashboard.

Module 7: Membership Churn Prediction Module

Purpose:

Identifies members who are likely to cancel or fail to renew their memberships using AI-based predictive analytics.

Key Features:

Attendance Monitoring

Renewal Analysis

Risk Classification

Retention Suggestions

Inputs:

Attendance History

Membership Expiration Date

Payment History

Outputs:

High-Risk Member List

Churn Prediction Report

Interaction with Other Modules:

Uses information from the Membership Management module and displays notifications on the Dashboard for administrator review.

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Module 8: Expense Leak Detector Module

Purpose:

Analyzes financial expenses and detects abnormal increases that may indicate unnecessary or excessive spending.

Key Features:

Expense Comparison

Monthly Trend Analysis

Automatic Alerts

Financial Recommendations

Inputs:

Historical Expense Records

Current Expense Data

Outputs:

Expense Alert Notifications

Expense Analysis Report

Interaction with Other Modules:

Uses data from the Expense Management module and provides alerts through the Dashboard and Financial Reports module.

Low-Fidelity Prototype Screens

Screen 1: Home Page

Functionality:

The Home Page serves as the main entry point of the system. It displays the gym's branding, system overview, and navigation options such as Login and About the System.

User Workflow:

User visits the website > Reads system information > Clicks Login.

Screen 2: Login Page

Functionality:

Allows administrators to securely log into the Smart Gym Finance Manager.

User Workflow:

Administrator enters username and password > System validates credentials > Administrator is redirected to the Dashboard.

Screen 3: Dashboard

Functionality:

Displays an overview of monthly revenue, total expenses, profit, membership statistics, AI notifications, and graphical reports.

User Workflow:

Administrator logs in > Reviews dashboard > Selects a module from the navigation menu.

Screen 4: Membership Management Page

Functionality:

Allows administrators to add, update, search, and manage member information, payment records, and membership renewals.

User Workflow:

Administrator selects Membership Module > Adds or edits member information > Saves changes.

Screen 5: Expense Management Page

Functionality:

Provides an interface for recording and monitoring operational expenses.

User Workflow:

Administrator enters expense details > Selects category > Saves expense > Expense is automatically added to financial records.

Screen 6: Financial Reports Page

Functionality:

Generates financial reports such as Revenue Reports, Expense Reports, and Profit and Loss Statements.

User Workflow:

Administrator selects report type > Chooses report period > System generates report > User exports report to PDF or Excel.

Screen 7: AI Analytics Page

Functionality:

Displays AI-generated Revenue Forecasts, Membership Churn Predictions, and Expense Leak Detection results.

User Workflow:

Administrator opens AI Analytics > Reviews predictions and alerts > Uses insights for financial planning.

Screen 8: User Profile and Settings Page

Functionality:

Allows administrators to update account information, change passwords, and manage system settings.

User Workflow:

Administrator opens Settings > Updates profile information > Saves changes.

Gantt Chart – Development Timeline (2-Week Schedule)

Project Phase	**Duration**
Requirements Gathering	Day 1
System Analysis and Design	Day 1 – Day 2
Prototype Development (Figma/Wireframes)	Day 2 – Day 3
Database Design	Day 3
Coding and Implementation	Day 4 – Day 10
AI Model Integration	Day 8 – Day 10
Testing and Debugging	Day 11 – Day 12
User Acceptance Testing	Day 13
Deployment	Day 13
Documentation	Day 1 – Day 14
Final Presentation and Submission	Day 14

Development Timeline Summary

Week 1

Day 1: Requirements Gathering

Day 1–2: System Analysis and Design

Day 2–3: Prototype Development

Day 3: Database Design

Day 4–7: Front-End and Back-End Development

Week 2

Day 8–10: AI Integration and System Completion

Day 11–12: Testing and Debugging

Day 13: User Acceptance Testing and Deployment

Day 14: Documentation Completion and Final Presentation

Development Sequence and Dependencies

The development of the Smart Gym Finance Manager begins with gathering system requirements from gym owners and administrators to determine the features and functionalities needed for effective financial management. After identifying the requirements, the project proceeds with system analysis and interface design, where the development team creates the system architecture, database structure, navigation flow, and user interface prototypes.

Once the design phase is completed, the database is developed before the coding phase begins. During implementation, the eight system modules are developed individually and integrated into a single cloud-based application. Artificial Intelligence features, including Revenue Forecasting, Membership Churn Prediction, and Expense Leak Detection, are integrated after the core financial modules have been completed.

Testing and debugging are then conducted to ensure that all modules function correctly and that system errors are resolved before deployment. User Acceptance Testing is performed with selected users to evaluate the usability, functionality, and overall performance of the system. After successful testing, the Smart Gym Finance Manager is deployed to the cloud environment, making it accessible through the web. Documentation is prepared throughout the development process and finalized before the project's final presentation and submission. Despite the two-week development schedule, the project remains achievable by utilizing cloud technologies, modern web development frameworks, and agile development practices.

Chapter IV: Division of Labor

Project Team Organization

The development of the Smart Gym Finance Manager requires teamwork and proper task distribution to ensure the successful completion of the project. Each member is assigned specific responsibilities based on their role and expertise. Through collaboration and effective communication, the team aims to develop a reliable cloud-based financial management system for fitness centers.

1. Calderon, Limar Glen – Project Leader

Responsibilities:

Lead and supervise the project development.

Coordinate team tasks and monitor project progress.

Oversee system integration and cloud deployment.

Assist in front-end and back-end development.

Lead the final project presentation.

2. David, Gerardo Jr. – Member

Responsibilities:

Assist in designing the user interface.

Develop the Membership Management Module.

Assist in the Expense Management Module.

Perform system testing and debugging.

Help prepare the project documentation.

3. Cabato, Aaron Ram – Member

Responsibilities:

Design and manage the project database.

Develop the Financial Reports Module.

Implement cloud database connectivity.

Assist in system integration and testing.

Ensure data security and backup.

4. Cabato, Justine Cyrus – Developer

Responsibilities:

Develop the AI Revenue Forecast Module.

Develop the Membership Churn Prediction Module.

Develop the Expense Leak Detector Module.

Assist in coding, testing, and debugging.

Prepare the technical documentation and user manual.

Development Collaboration

Throughout the project, all members will work together during system analysis, implementation, testing, documentation, and final presentation. Regular meetings will be conducted to monitor progress, discuss challenges, and ensure that all project objectives are achieved within the scheduled timeline.

Chapter Summary

This chapter presented the division of labor for the development of the Smart Gym Finance Manager. Each member has designated responsibilities that contribute to the successful completion of the project. Through teamwork, proper coordination, and shared responsibilities, the project team aims to develop a functional, efficient, and reliable cloud-based financial management system for fitness centers.